Vision of the Department

To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.

Mission of the Department

To nurture computer engineering professionals by offering: Dynamic learning environment that encourages problem-solving using emerging technology, · National Education Policy aligned evidence-based technical education, fostering innovation, research, and industry–institute collaboration, Holistic, ethical and sustainable development for lifelong success.

Session 2025-2026(EVEN)

Vision: Dream of where you want.	**Mission:** Means to achieve Vision

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	**Preparation**	**P: Preparation**	**Pep-CL abbreviation pronounce as Pep-si-lL easy to recall**
PEO2	**Core Competence**	**E: Environment (Learning Environment)**	
PEO3	**Breadth**	**P: Professionalism**	
PEO4	**Professionalism**	**C: Core Competence**	
PEO5	**Learning Environment**	**L: Breadth (Learning in diverse areas)**	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:
Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

"I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life." *to contribute to the development of cutting-edge technologies and Research*.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

SHEIKH WASIMUDDIN

Name and Signature of Student and Date
(Signature and Date in Handwritten)

Shiky
6/3/26

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Session	**2024-25 (EVEN)**	**Course Name**	**Mobile and Web Application for IOT Lab**
Semester	**6**	**Course Code**	**23IOT1624**
Roll No	**63**	**Name of Student**	**SHEIKH WASIMUDDIN**

Practical Number	**06**
Course Outcome	1. Understand and apply basic Kotlin programming constructs including variables, control structures, functions, null safety, and object-oriented concepts for Android app development. 2. Design and develop responsive user interfaces using XML layout files, ViewGroups, and widgets in Android Studio, adhering to Material Design principles. 3. Implement event handling and application logic using Kotlin to capture user input, manage app state, navigate between screens, and interact with Android system components like Toasts and Dialogs 4. Integrate learned concepts to develop a mini Android application that demonstrates use of ViewModel, multi-screen navigation, and proper UI/UX design for real-world problem-solving
Aim	Enhance UI and ensure responsiveness
Problem Definition	ImageView, ScrollView, ListView, responsive layouts, emulator and device testing.
Theory (100 words)	User Interface (UI) design is an essential part of Android application development because it directly affects how users interact with the application. A well-designed interface makes the app visually appealing, easy to navigate, and comfortable to use. To enhance the UI and improve usability, Android provides several layout components and views such as **ImageView, ScrollView, and ListView**. These components help developers present information in a structured and user-friendly way. **ImageView** is used to display images like icons, logos, or illustrations within the application. It improves the visual appearance of the interface and helps communicate information more clearly. **ScrollView** allows the user to scroll through content when the layout contains more elements than can fit on the screen. This ensures that all content remains accessible without making the interface cluttered. **ListView** is used to display multiple items in a vertical list format. It is useful for showing data such as names, options, or records in an organized manner.

	Along with these components, creating **responsive layouts** is important so that the application adjusts properly to different screen sizes and resolutions. Finally, testing the application on both **Android emulators and real devices** is necessary to ensure the layout behaves correctly. Proper testing helps identify UI issues and ensures smooth performance across various devices and screen configurations.
Procedure and Execution (100 Words)	**Step for Implementation:** First, a new Android project was created in Android Studio. The layout file **activity_main.xml** was designed to improve the user interface using different UI components. An **ImageView** was added to display the application logo or image at the top of the screen. A **ScrollView** was used to make the layout scrollable so that all UI elements remain accessible even on smaller screens. A **ListView** was then included to display multiple items in a structured list format. Responsive layout properties such as proper margins, padding, and match_parent or wrap_content were applied. Finally, the application was executed and tested on both the Android emulator and a physical device. **Step for Implementation:** 1. Open **Android Studio** and create a new Android project. 2. Navigate to the **res → layout → activity_main.xml** file. 3. Add an **ImageView** to display an image or logo. 4. Wrap the layout inside a **ScrollView** to enable scrolling. 5. Add a **ListView** to display multiple items in list format. 6. Adjust margins, padding, and layout sizes for responsiveness. 7. Write the required logic in **MainActivity.java / MainActivity.kt** if needed. 8. Run the application using the **Android emulator or a real device** to test the UI.
	Code: **1. Manifest** <?xml version="1.0" encoding="utf-8"?> <manifest xmlns:android="http://schemas.android.com/apk/res/android" xmlns:tools="http://schemas.android.com/tools"> <application android:allowBackup="true" android:dataExtractionRules="@xml/data_extraction_rules" android:fullBackupContent="@xml/backup_rules" android:icon="@mipmap/ic_launcher" android:label="@string/app_name"

```
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.Practical5_iot63">
        <activity
            android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />

                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
    </application>

</manifest>
```

2. MainActivity.kt

```
package com.example.practical6_iot63

import androidx.appcompat.app.AppCompatActivity
import android.os.Bundle
import androidx.recyclerview.widget.LinearLayoutManager
import androidx.recyclerview.widget.RecyclerView

class MainActivity : AppCompatActivity() {

    lateinit var recyclerView: RecyclerView

    val languages = arrayOf(
        "Hindi",
        "English",
        "Bengali",
        "Telugu",
        "Marathi",
        "Tamil",
        "Urdu",
        "Gujarati",
        "Kannada",
        "Odia",
        "Malayalam",
        "Punjabi",
        "Assamese",
        "Maithili",
        "Sanskrit",
```

```
        "Konkani",
        "Manipuri",
        "Bodo",
        "Dogri",
        "Santali",
        "Kashmiri",
        "Sindhi",
        "Nepali"
    )

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)

        setContentView(R.layout.activity_main)

        recyclerView = findViewById(R.id.recyclerView)

        recyclerView.layoutManager = LinearLayoutManager(this)

        val adapter = LanguageAdapter(languages)

        recyclerView.adapter = adapter
    }
}
```

3. ActivityMain.xml

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="10dp"
    android:layout_marginTop="40dp">

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Programming Languages By Tanmay"
        android:textSize="22sp"
        android:textStyle="bold"
        android:gravity="center"
        android:paddingBottom="10dp"/>
```

```
    <ListView
        android:id="@+id/listView"
        android:layout_width="match_parent"
        android:layout_height="match_parent"/>

</LinearLayout>
```

4. LanguageAdaptor.kt

```
package com.example.practical6_iot63

import android.view.LayoutInflater
import android.view.View
import android.view.ViewGroup
import android.widget.TextView
import androidx.recyclerview.widget.RecyclerView

class LanguageAdapter(private val languages: Array<String>) :
    RecyclerView.Adapter<LanguageAdapter.ViewHolder>() {

    class ViewHolder(itemView: View) :
RecyclerView.ViewHolder(itemView) {
        val text: TextView = itemView.findViewById(R.id.langName)
    }

    override fun onCreateViewHolder(parent: ViewGroup, viewType: Int):
ViewHolder {
        val view = LayoutInflater.from(parent.context)
            .inflate(R.layout.list_item, parent, false)

        return ViewHolder(view)
    }

    override fun onBindViewHolder(holder: ViewHolder, position: Int) {
        holder.text.text = languages[position]
    }

    override fun getItemCount(): Int {
        return languages.size
    }
}
```

5. **List_item.xml**

```
    <TextView
    xmlns:android="http://schemas.android.com/apk/res/android"
```

	android:id="@+id/langName" android:layout_width="match_parent" android:layout_height="wrap_content" android:textSize="18sp" android:padding="16dp" android:layout_marginBottom="4dp"/>
	Output:

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Indian Languages By Sheikh

Odia

Malayalam

Punjabi

Assamese

Maithili

Sanskrit

Konkani

Manipuri

Bodo

Dogri

Santali

Kashmiri

Sindhi

Nepali

Output Analysis	The developed Android application successfully demonstrated an improved and responsive user interface. The **ImageView** displayed the image clearly at the top of the screen, enhancing the visual appearance of the application. The **ScrollView** allowed users to scroll through the content smoothly when the layout exceeded the screen size, ensuring that all elements remained accessible. The **ListView** effectively displayed multiple items in a structured vertical list, making the information easy to read and navigate. The layout adjusted properly on different screen sizes, confirming its responsiveness. Testing on both the emulator and a physical device showed that the application functioned correctly without layout distortion or usability issues.
Link of student Github profile where lab assignment has been uploaded	https://github.com/sheikhwasimuddin
Conclusion	In conclusion, the experiment helped in understanding how different Android UI components can be used to create a responsive and user-friendly interface. By implementing elements such as **ImageView, ScrollView, and ListView**, the application layout became more organized and visually appealing. The use of responsive design techniques ensured that the interface adapted properly to different screen sizes and resolutions. Testing the application on both an emulator and a real device confirmed that the UI components worked smoothly and displayed content correctly. Overall, this activity provided practical knowledge about improving application usability and designing efficient Android layouts for better user interaction and experience.

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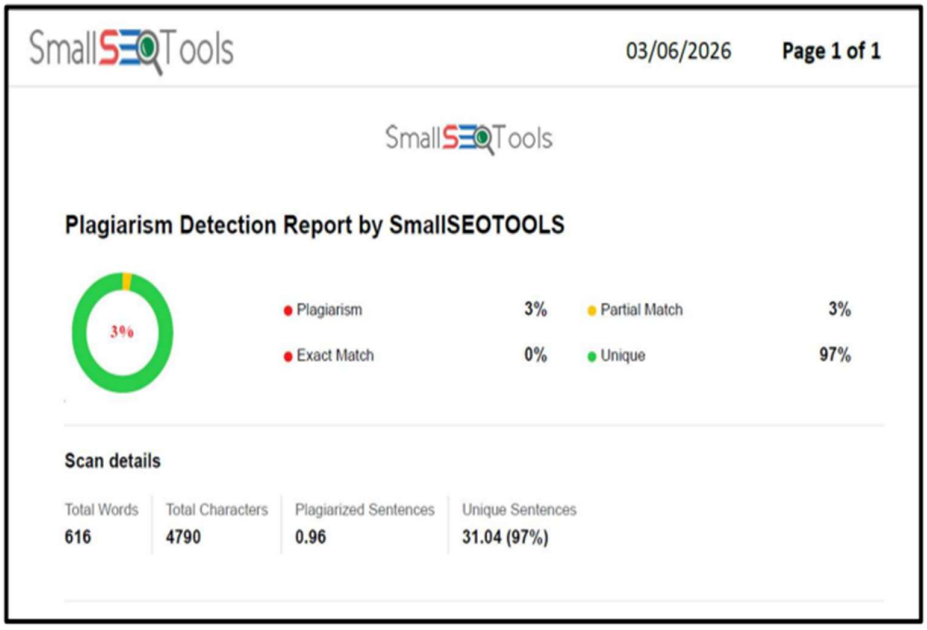
Plag Report (Similarity index < 12%)	
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